

VANIER COLLEGE MUSIC DEPARTMENT

Faculty of Arts, Business and Social Sciences

Sound Design for Multimedia (551-301-VA) • Cook

HANDOUT: Overview of Sound Design in Media

What is Sound Design?

Sound design is the art and practice of creating and manipulating audio to support storytelling in media. It shapes the emotional tone, atmosphere, and realism of a project.

Sound design may include:

- Creating original sounds (e.g. sci-fi effects, creature roars)
- Recording dialogue, Foley, and ambience
- Manipulating audio with EQ, reverb, pitch shifting, and other effects
- Integrating sound seamlessly with visuals to enhance the audience experience

Key Elements of Sound Design

- Dialogue – Spoken word; may include ADR (Automated Dialogue Replacement)
- Foley – Re-creation of everyday sounds (footsteps, clothing rustle)
- Ambience – Background environmental sound that sets the mood
- Sound Effects (SFX) – Non-musical effects that highlight action or events
- Music – Score or soundtrack to add emotion and rhythm
- Silence – A powerful tool to build tension or draw focus

Where Sound Design is Used

- Film & TV – Enhances realism and immersion
- Video Games – Interactive sound that reacts to gameplay
- Theatre – Live soundscapes and cue-based effects
- Animation – Entire sound worlds built from scratch
- Podcasts & Radio – Storytelling through audio alone
- Advertising – Catching attention and reinforcing brand identity

Why Sound Design Matters

- Enhances Emotion – Guides audience feelings
- Builds Immersion – Makes fictional worlds believable
- Directs Attention – Focuses on important details in a scene
- Strengthens Storytelling – Adds depth, meaning, and mood

HANDOUT: The General Characteristics of Sound

Simple Harmonic Motion & Sound

- Sound is caused by vibrations creating fluctuations in air pressure.
- Vibrating objects (e.g., tuning forks) push and pull air molecules, forming pressure waves.
- Pressure changes create a sinusoidal (sine wave) pattern over time.
- Amplitude = the size of pressure change from normal atmospheric pressure.
- **Key terms:**
 - **Equilibrium:** Rest position, no pressure
 - **Crest:** High point of a wave
 - **Trough:** Low point of a wave
 - **Wavelength:** Distance between identical points in successive waves
 - **Frequency:** Waves per second
 - **Amplitude:** Max distance from equilibrium (loudness)
 - **Period:** Time for one complete cycle

Hearing Requirements

- Fluctuations must be strong enough to move the eardrum but not harmful.
- Must occur at rates between ~20 Hz and 20,000 Hz (human hearing range).
- **Pitch:** Perceived frequency (higher frequency = higher pitch).
- **Hertz (Hz):** Unit of frequency (cycles per second).
- One octave = frequency ratio of 2:1.

Complex Tones & Timbre

- Most sounds are **complex tones** (multiple frequencies).
- **Fundamental** = lowest frequency; **overtones** = higher frequencies.
- Harmonic overtones are multiples of the fundamental; inharmonic are not.
- Timbre = unique sound quality determined by a mix of harmonic and inharmonic overtones.
- Sounds range from pure sine waves to random white noise.
- Overtones change over time, shaping the character of sound.

Amplitude & Loudness

- Loudness is perceived based on amplitude ratio, not exact difference.
- Sensitivity peaks between 300–7,000 Hz (speech range).
- Loudness measured in **decibels (dB)** (logarithmic scale).
- Thresholds:
 - 0 dB = hearing threshold

- 130 dB = pain threshold
- Whisper ~15–25 dB, Normal speech ~65 dB, Rock concert ~120 dB

Amplitude Envelope

- Describes how loudness changes over time.
- **Attack** = rise from silence to peak.
- Envelope helps us recognize and distinguish sounds.

Signal-to-Noise Ratio

- Ratio of wanted sound to background noise.
- CDs achieve ~96 dB signal-to-noise ratio.

Digital Representation of Sound

- Works like film: many samples per second create illusion of continuity.
- **Sampling rate**: number of samples per second (CD = 44.1 kHz).
- **Bit depth**: sample size, affecting dynamic range.
- Higher rates/depths = better quality.
- Minimum standard: 44.1 kHz, 16-bit; preferred: 96 kHz, 24-bit.
- **ADC** converts analog signals to digital; **DAC** converts back to analog.
- Storage formats:
 - **Uncompressed** (WAV, AIFF) = best quality, large size
 - **Compressed** (MP3, WMA) = smaller size, some quality loss

Clipping

- Happens when amplitude exceeds recording system's max level.
- Causes distortion or harsh noise.
- Should be avoided to preserve sound quality.

Audio Formats

- **Analog**: Being phased out.
- **Digital**: Flexible, easy to edit, no tape cost, no generation loss.
- **Software examples**: Pro Tools, Logic, Cubase.

HANDOUT: Creating an audio file in **Adobe Audition**, from start to export:

1. Open Adobe Audition

2. Create a New File

- Go to **File > New > Audio File**
- Name your file
- Choose sample rate (e.g., 44100 Hz for music, 48000 Hz for video)
- Set channels (Mono or Stereo)
- Choose bit depth (16-bit for CD quality, 24-bit for higher fidelity)

3. Record or Import Audio

- To record: Arm input, check microphone settings, then press the **Record button**
- To import: Go to **File > Import > File**, then drag the file into the editor or multitrack session

4. Edit the Audio

- Trim unwanted parts
- Use **Fade In/Out** to smooth beginnings and endings
- Normalize audio for consistent levels
- Apply EQ, compression, or noise reduction if needed

5. Work in Multitrack (if using multiple audio sources)

- Create a **New Multitrack Session** under **File > New > Multitrack Session**
- Drag audio files to different tracks
- Adjust timing, crossfades, and volume automation

6. Mix the Audio

- Balance track levels
- Pan left/right for stereo image
- Apply effects on individual tracks or the master bus

7. Master the File

- Add final compression, EQ, and limiting on the master track
- Check overall loudness and clarity

8. Preview Before Export

- Play through the entire project to catch clipping, noise, or mistakes

9. Export the Audio File

- Go to **File > Export > File** (for a single audio file) or **File > Export > Multitrack Mixdown > Entire Session** (if in multitrack)
- Choose format (WAV for high quality, MP3 for compressed, etc.)
- Set output sample rate, bit depth, and channels
- Select export destination
- Click **OK/Export**

HANDOUT: Choosing Proper Bit Depth & Sample Rate

What Is Sample Rate & Bit Depth?

- **Sample Rate:** The number of audio snapshots captured per second.
 - Higher rates = more detail, larger files.
- **Bit Depth:** How much volume range (dynamic range) is captured.
 - Higher depths = cleaner sound with less noise.

Sample Rate Guide

Sample Rate	Best For	Notes
44.1 kHz	Music, podcasts	Standard for CDs & streaming
48 kHz	Video, film, broadcast	Industry standard for video & TV
96 kHz	High-end audio	More detail; bigger files
192 kHz	Archival work	Rarely needed; very large files

Bit Depth Guide

Bit Depth	Best For	Notes
16-bit	Final tracks, podcasts	CD-quality; good for delivery
24-bit	Recording & editing	More headroom; prevents clipping
32-bit float	Advanced mixing	Huge dynamic range; large files

Setting in Adobe Audition

- **When creating a new file**
 - Go to File → New → Audio File
 - Choose Sample Rate and Bit Depth
- **When recording**
 - Open Edit → Preferences → Audio Hardware
 - Confirm input/output device supports your chosen rate
 - Match settings in File → New → Audio File

- **When exporting**
 - Go to File → Export → File...
 - Choose Format Settings (e.g., 16-bit 44.1 kHz WAV for delivery)

Rules of Thumb

- Podcasts & Music: 44.1 kHz, 24-bit (recording), export 16-bit
- Video Projects: 48 kHz, 24-bit (recording), export 16-bit or 24-bit
- Sound Design: 48 or 96 kHz, 24-bit or 32-bit float
- Final Delivery: 44.1 or 48 kHz, 16-bit

Tip: In most real-world projects, **44.1 or 48 kHz at 24-bit** is the sweet spot.

HANDOUT: Common Processing Steps for Professional Sound

Editing & Cleanup

- **Trim** unnecessary silence or noise at the beginning and end.
- **Remove clicks, pops, and hums** (with tools like iZotope RX or Adobe Audition's noise reduction).
- **Comping & arrangement:** choose the best takes and organize clips.

Noise Reduction

- Use **noise reduction** or a **noise gate** to cut background hiss, hum, or room noise.
- Be subtle — overdoing it can make audio sound unnatural.

Equalization (EQ)

- **High-pass filter** to remove unnecessary low frequencies (rumble, mic handling noise).
- **Boost clarity** (around 2–5 kHz for voice, 5–10 kHz for brightness in instruments).
- **Cut muddiness** (200–500 Hz for voice, 150–250 Hz for many instruments).
- Apply **gentle boosts/cuts** instead of extreme EQing.

Dynamics Control

- **Compression** to even out loud and quiet parts:
 - Ratio: 2:1–4:1 for voice/instruments.
 - Attack/Release: Adjust for natural feel.
- **De-essing** for vocals to reduce harsh “S” sounds.
- Sometimes **parallel compression** to add body without losing dynamics.

Enhancement (Optional)

- **Saturation** or **subtle distortion** for warmth and richness.
- **Exciter / Harmonic enhancer** for brightness and clarity.

Spatial Effects

- **Reverb** for natural space and depth — avoid making it too “roomy.”
- **Delay** if stylistically appropriate (e.g. for vocals, instruments).
- Keep effects subtle so the sound remains clear.

Balancing & Automation

- Adjust **levels** so the sound sits well in the mix.
- Use **automation** for volume, panning, or effect changes throughout the track.

Stereo Imaging

- Use **panning** and **stereo widening** carefully to give space, but don't lose mono compatibility.

Final Touches

- **Limiter** on the master to prevent clipping.
- **Reference check:** Compare to professional recordings on good speakers and headphones.
- **Export at proper settings** (e.g. 24-bit, 48kHz for video, or 16-bit, 44.1kHz for music distribution).

HANDOUT: Equalizer Basics

What is an equalizer?

An **equalizer (EQ)** is an audio tool that lets you adjust the balance of different frequency ranges in a sound.

Think of it like a very precise volume control, but instead of changing the overall loudness, it changes how loud or quiet specific parts of the frequency spectrum are—such as bass, midrange, or treble.

In practice:

- **Low frequencies** (bass) → rumble, warmth, or thump.
- **Mid frequencies** → clarity of voices and instruments.
- **High frequencies** (treble) → brightness, air, or sizzle.

Uses in sound design and mixing:

- Removing unwanted noise (e.g., low hum or harsh hiss).
- Making a voice or instrument stand out in a mix.
- Shaping the “tone” of a sound to match a style or mood.

If you picture sound like a photograph, EQ is a bit like adjusting the brightness in specific colors—so you can make some elements pop and others fade.

Specific Frequency Ranges

- **Low Bass (40–80 Hz)**
 - Adds *power* and *feeling*.
 - Too much → muddy sound, especially on large speakers.
 - Common in hip hop, rap, and dance music.
 - 20–40 Hz rarely used except for pipe organs.
- **Bass (80–250 Hz)**
 - Determines *fatness* and *fullness*.
 - 100 Hz: adds body to guitars and bass (but can sound boomy).
 - 200 Hz: adds vocal fullness; reducing can improve clarity.
 - 80 Hz boost enhances kick drum low end.
- **Low Mid (250–500 Hz)**
 - Adds clarity to bass and lower register instruments.
 - Too much → muffled instruments, “cardboard box” drum sound.
 - Balancing kick and bass here improves separation.
 - Often reduced on overheads and toms for clarity.
- **Mid Range (500 Hz–2 kHz)**

- 500–1kHz: horn-like quality.
- 1–2kHz: tinny, “telephone” sound.
- 800 Hz: presence/reduces nasal tone in vocals.
- 1.5 kHz: adds attack to bass and rhythm guitars.
- **High Mid (2–6 kHz)**
 - Controls attack and projection.
 - 2.5 kHz: softer beater kick drum attack.
 - 4 kHz: harder beater kick drum attack.
 - Boost adds projection to vocals and distinction to guitars.
 - Too much → harshness and listening fatigue.
- **High/Treble (6–20 kHz)**
 - Adds brilliance and “air.”
 - 7 kHz: “S” sounds in vocals (use carefully).
 - 15 kHz+: breathiness in vocals; cymbal “sizzle.”
 - 10 kHz+: general brightness.

Types of Equalizers

- **Graphic Equalizer**
 - Uses sliders; visually shows frequency response.
 - Commonly divided into octaves (7–9 bands), $\frac{1}{2}$ octaves (15 bands), or $\frac{1}{3}$ octaves (31 bands).
 - 31-band EQ covers 20 Hz to 20 kHz.
 - Sliders typically adjust ± 6 dB or ± 12 dB.
- **Parametric Equalizer**
 - Uses knobs (no visual graph).
 - Typically adjusts 3–5 frequencies per unit.
 - Fully controls:
 - Center Frequency
 - Amplitude (Gain)
 - Bandwidth (Q) → high Q = narrow, low Q = wide

Other EQ Types

- **High Shelf:** boosts/cuts frequencies *above* set frequency.
- **Low Shelf:** boosts/cuts frequencies *below* set frequency.
- **Peak EQ:** boosts/cuts at set frequency and nearby range.
- **High Cut Filter (Low Pass):** reduces frequencies above cutoff; slope adjustable.
- **Low Cut Filter (High Pass):** reduces frequencies below cutoff; slope adjustable.

HANDOUT: Compressors, etc

Compressor

- Reduces signal levels above a set threshold by a set ratio.
- Threshold: level where compression begins.
- Ratio: how much the signal is reduced (e.g., 2:1 means 10dB becomes 5dB over threshold; 4:1 means 12dB becomes 3dB over threshold, etc).
- Attack: time compressor takes to react once threshold is exceeded.
- Release: time to return to normal after the signal drops below threshold.
- Output Gain (Make-up Gain): restores level lost during compression.
- Hard Knee: sudden compression at threshold.
- Soft Knee: gradual compression around threshold for smoother effect.
- Side-chaining: uses one track to control compression on another (e.g., ducking, pumping).
- Tip: Route the entire signal through the compressor (not like an aux effect).

Limiter

- Extreme form of compression; prevents signals from exceeding threshold.
- Acts at very high ratios (often infinity:1).
- Some allow slight increases above threshold; “brick wall” limiters allow none.
- Essential for digital audio (prevents clipping above 0 VU).
- Protects live sound systems from feedback and damage.

Expander

- Opposite of compression; reduces levels below a set threshold.
- Threshold: level where expansion starts.
- Ratio: sets how much signal below threshold is reduced (e.g., 1:3 reduces 10dB below threshold to 30dB).

Noise Gate

- Lets signal through when above threshold; cuts off when below.
- Useful for removing background noise in silent sections.
- Works as an expander on low-level signals.
- Release Time:
 - Too long → noise reduction is obvious.
 - Too short → cuts off reverb and natural decay.

HANDOUT: Common Audio Effects

Reverb

- Created by multiple reflections of sound in a room, arriving after the direct sound.
- Early reflections: directional, linked to room size/shape.
- Late reflections (diffuse reverberation): random, define perceived “room size.”
- Spaciousness enhanced by incoherent signals at each ear (high ceilings help).
- Reflective surfaces increase reverb; absorptive materials and people reduce it.
- Excess reverb reduces speech/music clarity.
- In recording, reverb often added artificially for control.
- **Types:**
 - *Gated reverb*: abruptly stops after set time.
 - *Reverse reverb*: sound grows louder before cutting off.
- **Common parameters**: pre-delay, room size, reverb time, mix, high-frequency damp, gate time, gate threshold.

Delay

- Splits and delays signal, then recombines it, often with feedback to create echoes.
- **Types:**
 1. Single delay – one echo.
 2. Multi-tap delay – multiple echoes with separate times/decays.
 3. Ping-pong delay – alternates echoes between stereo channels.
- **Parameters**: delay time, feedback, decay level, mix, tempo sync.

Chorus

- Duplicates signal, delays slightly, modulates pitch up/down, then combines with original to create “multiple” voices.

Phase Shifter (Phaser)

- Duplicates signal, shifts phase, combines with original to create moving frequency notches.

Flanger

- Similar to phaser but uses short, varying delay (~20 ms) with feedback, producing “jet plane” swoosh.
- Originated from slowing one reel-to-reel tape reel.

Vibrato

- Modulates pitch only; parameters: rate (speed), depth (amount).

- Like chorus without the original dry sound.

Tremolo

- Modulates volume only; parameters: rate (speed), depth (intensity).

HANDOUT: Professional Sound Processing Checklist

Edit & Clean

- Trim silence and unwanted parts
- Remove clicks, pops, and hums
- Organize and arrange best takes

Noise Reduction

- Reduce background hiss, hum, or room noise
- Apply gently to avoid artifacts

Equalization (EQ)

- Roll off unnecessary low frequencies
- Reduce muddiness in the low-mid range
- Add clarity and brightness where needed

Dynamics Control

- Use compression to smooth out volume
- Apply de-essing if needed for harsh “S” sounds
- Keep natural dynamics intact

Enhancement (Optional)

- Add light saturation for warmth
- Use an exciter to brighten the sound

Spatial Effects

- Add subtle reverb for space and depth
- Use delay only if stylistically appropriate

Balance & Automation

- Adjust levels for clarity in the mix
- Use automation to control changes over time

Stereo Imaging

- Pan elements for space and clarity
- Use stereo widening sparingly

Final Touches

- Apply a limiter to prevent clipping

- Compare to professional references
- Export at proper settings (e.g., 24-bit/48kHz)

HANDOUT: Microphones

Microphone Pickup Patterns & Types

1. Omnidirectional

- Picks up sound **equally from all directions**
- Great for **natural room sound** and **group vocals**
- More **forgiving** with mic placement/angle
- Risk: can capture **too much ambient noise** if not positioned well

2. Unidirectional

- Sensitive to sound from **one main direction**
- **Cardioid pattern** rejects sound from behind
 - Reduces bleed (e.g., guitar amp near drum kit)
- **Supercardioid** = narrower pattern, more isolation
- Useful in **crowded setups** with many mics close together

3. Shotgun

- **Most directional** microphone type
- Captures **direct sound** while rejecting room/reverb
- Important when mic is **far from sound source**
- Usually **more expensive** for good sound quality

Microphone Technologies

1. Dynamic Microphones

- Work like a **reverse speaker** (diaphragm creates electricity from sound waves)
- **Rugged** and handle **high sound pressure levels**
 - Good for **kick drums, snares, loud guitar amps, aggressive vocals**
- **Lower cost** and **durable**

2. Condenser Microphones

- **More sensitive**; capture fine detail and high frequencies
- Great for **acoustic guitar, cymbals, and vocals**
- Require **phantom power** (48V) via XLR cable
- Types:
 - **Large diaphragm** → more sensitive, better for **vocals**, more expensive
 - **Small diaphragm** → less sensitive, still detailed

HANDOUT: Podcast Recording Workflow in Adobe Audition

1. Prepare Your Session

- Open Adobe Audition and choose Multitrack Session (File > New > Multitrack Session).
- Name your session (e.g., Podcast_Episode_01), choose save location.
- Set Sample Rate: 44.1 kHz (standard) or 48 kHz, Bit Depth: 24-bit.

2. Set Up Your Tracks

- Create one mono track per microphone.
- Assign correct mic or interface channel to each track's Input dropdown.
- Name each track (e.g., Host, Guest 1, Guest 2).

3. Configure Audio Hardware

- Go to Edit > Preferences > Audio Hardware.
- Device Class: ASIO (Windows) or Core Audio (Mac).
- Select audio interface for Input and Output.
- Set low latency without distortion.

4. Check Mic Levels

- Speak into each mic and watch meters.
- Target: -12 dB to -6 dB peaks.
- Adjust gain on audio interface.

5. Arm Tracks for Recording

- Click red R on each track.
- Enable Input Monitoring (I) if needed.
- Do a short test recording.

6. Record Your Podcast

- Press main Record button.
- Keep consistent mic distance, speak clearly.
- If mistake happens, pause and restart.

7. Save and Back Up

- Stop recording and save session.
- Back up .sesx file and audio folder.

8. Basic Editing

- Trim silences and mistakes.
- Use Razor Tool (R) to cut, Move Tool (V) to arrange.
- Apply noise reduction if needed.

9. Apply Processing

- EQ: roll off below 80 Hz, boost 3–5 kHz for clarity.
- Compression: even out levels.
- Limiter: prevent peaks from clipping.

10. Export Your Podcast

- File > Export > Multitrack Mixdown > Entire Session.
- Save as MP3 (192 kbps) or WAV for archive.
- Add metadata (title, episode, host).

HANDOUT: Quick Podcast Recording Checklist – Adobe Audition

Before Recording

- New Multitrack Session (44.1 kHz or 48 kHz, 24-bit)
- One mono track per mic, set inputs
- Name each track
- Check Audio Hardware settings
- Test mic levels (-12 to -6 dB peaks)

During Recording

- Arm tracks for recording (R)
- Enable monitoring (I) if needed
- Do short test recording
- Maintain mic distance & clarity
- If mistake, pause & restart

After Recording

- Save and back up session
- Trim mistakes & silences
- Apply EQ, compression, limiter
- Export MP3 (192 kbps) or WAV
- Add metadata

HANDOUT: ADR Workflow in Adobe Audition

1. Prepare Your Session

- Open Adobe Audition and choose Multitrack Session (File > New > Multitrack Session).
- Name the session (e.g., Scene_03_ADR), set the Sample Rate to match your video (48 kHz for film/TV), and pick a save location.
- Import your video and production audio (File > Import > File...).
- Drag the video onto a track — Audition will open the Video panel automatically.
- Place your original audio (with problematic dialogue) on a reference track.

2. Set Up for Recording

- Create a new mono track for your ADR mic (Multitrack > Track > Mono Audio Track).
- Assign your input source (mic or audio interface channel) in the track's Input dropdown.
- Arm the track for recording (red R icon).
- Enable monitoring (I icon) so the actor can hear themselves through headphones without feedback.

3. Mark the Cues

- Play through the video and find the lines to replace.
- Place markers (M key) at the start of each cue.
- Name markers with the line of dialogue or take number for organization.

4. Cue the Talent

- Loop playback for the section (Transport > Loop Playback).
- Use a 3-beep or cue tone so the actor knows when to start.
- Optionally count in during practice runs.

5. Record the ADR

- Have the actor watch the video in sync and hit record.
- Perform multiple takes per line.
- Label each clip with take number.

6. Edit & Align

- Use Time Selection and Slip tools to nudge takes into sync with lip movements.
- Use waveform matching by lining up visual peaks.
- Trim unwanted noise from clip edges.

7. Match the Sound

- Apply EQ to match tone to surrounding audio.
- Add subtle reverb to match room ambience.
- Adjust gain to blend levels.

8. Export the ADR

- Mute reference track with bad dialogue.
- Export mix (File > Export > Multitrack Mixdown > Entire Session) or just ADR stem.
- Save as WAV (48 kHz, 24-bit) for delivery.

HANDOUT: Quick ADR Checklist – Adobe Audition

Before Recording

- Create New Multitrack Session (48 kHz sample rate)
- Import video and reference audio
- Create mono ADR track and set mic input
- Arm track for recording (R) & enable input monitoring (I)
- Place markers for each cue

During Recording

- Loop playback of cue
- Provide 3-beep or count-in before line
- Watch video for lip sync, perform multiple takes
- Label each clip with take number

After Recording

- Nudge clips to match lip movements
- Match tone & ambience (EQ + subtle reverb)
- Adjust gain to match surrounding dialogue
- Mute bad dialogue & export mix or stem